Prescribed-h_0 Intertwining through the Physical Archive

Exact Commutator Defects, Direct Slicing, and Sharp De-Aliasing Costs

Liang Wang

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Abstract

A fixed-shift hard packet cannot be recovered from a frozen or averaged family merely by retaining the same notation h_0. We give two exact and disjoint invocation routes. For a shift-tagged archive with stages T_j, prescribed-shift selectors P_j, and declared sliced stages $T_j^{(0)}$, define

$$D_j = P_j T_j - T_j^{(0)} P_{j-1}.$$

The full slicing defect is

$$P_s T_s \cdots T_1 - T_s^{(0)} \cdots T_1^{(0)} P_0 = \sum_{j=1}^{s} T_s^{(0)} \cdots T_{j+1}^{(0)} D_j T_{j-1} \cdots T_1.$$

Thus stagewise commutation is a proof-carrying sufficient condition for exact direct slicing, while a nonzero composite defect on the literal packet stops that declared route. If only averaged observations are available, the sharp squared evaluation cost is

$$K_X(h_0) = b(h_0)^* (A^* \Omega A)^\dagger b(h_0),$$

finite exactly when $b(h_0) \in \operatorname{ran}(A^* \Omega A)$. An averaged amplitude saving σ and $K_X(h_0) \le X^{2\lambda_{\mathrm{loc}} + o(1)}$ yield at most the fixed-shift saving $\sigma - \lambda_{\mathrm{loc}}$. Direct slicing and de-aliasing are not additive evidence for the same operation. The identities are L0, and their attachment to the TPC-123/124 archive is conditional L1. The actual growing shift archive and profile space are absent, so the interface remains NOT-TESTABLE. No L2 arithmetic estimate or prime-pair theorem is proved.

1 Two routes that must not be conflated

The original hard packet is defined at one prescribed nonzero shift h_0. TPC-119 therefore requires later operations to retain that shift, while TPC-115 shows that evaluation at one shift can have a nontrivial or infinite cost after averaging [3, 4]. TPC-123 and TPC-124 provide the appropriate archive and physical-reassembly schemas [8, 9]. They still leave two logically different possibilities:

(i) *direct slicing*: every stage preserves shift labels and the h_0 subspace is selected exactly;

(ii) *de-aliasing*: only averaged observations are bounded, and fixed evaluation is reconstructed from the actual profile space.

If the first route holds, there is no localization inversion. If only the second route is available, its Christoffel cost must be charged. Neither route proves an arithmetic bound.

Definition 1.1 (Claim levels)**.** Finite operator and pseudoinverse identities are L0. Their verified attachment to the literal fixed-h_0 packet is L1. A growing coefficient-specific fixed-shift power saving is L2.

2 Shift selectors and local defects

For $0 \le j \le s$, let E_j be the full shift-tagged record space at archive stage j, and let $E_j^{(0)}$ be the declared prescribed-h_0 record space. Let

$$P_j : E_j \to E_j^{(0)}$$

be coordinate selection and

$$T_j : E_{j-1} \to E_j, \qquad T_j^{(0)} : E_{j-1}^{(0)} \to E_j^{(0)}$$

be the full and declared sliced stages.

Definition 2.1 (Stage shift defect)**.** The j-th intertwining defect is

$$D_j = P_j T_j - T_j^{(0)} P_{j-1}. \tag{1}$$

The equality $D_j = 0$ says more than equality of two scalar outputs: it says that selection can be moved through the stage for every input column. A path implementation may check this entrywise by verifying that every nonzero edge keeps its shift tag.

3 The exact composite-defect telescope

Theorem 3.1 (Prescribed-shift intertwining telescope)**.** *For arbitrary compatible stage maps,*

$$\boxed{P_s T_s \cdots T_1 - T_s^{(0)} \cdots T_1^{(0)} P_0 = \sum_{j=1}^{s} T_s^{(0)} \cdots T_{j+1}^{(0)} D_j T_{j-1} \cdots T_1.} \tag{2}$$

Empty products are identities.

Proof. Insert the intermediate composites

$$T_s^{(0)} \cdots T_{j+1}^{(0)} P_j T_j \cdots T_1$$

for $j = s, s-1, \dots, 0$. The difference between consecutive terms is exactly

$$T_s^{(0)} \cdots T_{j+1}^{(0)} (P_j T_j - T_j^{(0)} P_{j-1}) T_{j-1} \cdots T_1.$$

Summing telescopes to (2). □

Corollary 3.2 (Stagewise sufficient certificate)**.** *If $D_j = 0$ for all j, then*

$$P_s T_s \cdots T_1 = T_s^{(0)} \cdots T_1^{(0)} P_0.$$

The direct h_0 slice then intertwines coefficientwise with the full archive.

Corollary 3.3 (Actual-packet stop witness)**.** *Let*

$$\Delta_X = P_s T_s \cdots T_1 - T_s^{(0)} \cdots T_1^{(0)} P_0.$$

If $\Delta_X c_X \neq 0$, the declared sliced chain does not reproduce the direct fixed-h_0 output on the literal packet c_X.

Remark 3.4 (Operator and one-vector quantifiers). $\Delta_X c_X = 0$ verifies one literal vector. The stronger archive identity $\Delta_X = 0$ verifies every native atom and is the appropriate coefficientwise H8 interface. Conversely, nonzero local D_j may cancel in the sum (2); it stops a claimed stagewise intertwining proof but does not alone prove $\Delta_X c_X \neq 0$.

4 A minimal cross-shift leakage example

Proposition 4.1 (Frozen notation does not preserve a shift)**.** *There is a two-shift stage whose selected output contains a coefficient from the other shift, even though the declared fixed-shift submatrix is unchanged.*

Proof. Let

$$P = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad T^{(0)} = (1).$$

Then

$$PT - T^{(0)}P = \begin{pmatrix} 0 & 1 \end{pmatrix} \neq 0.$$

The second source coordinate leaks into the selected output. Renaming that output "h_0" does not remove the defect. □

5 Sharp de-aliasing on an actual profile space

Suppose direct slicing is unavailable and an estimate controls only observations of a finite-dimensional profile $f \in V_X$. Let

$$A : V_X \to Y_X, \qquad \Omega = \Omega^* > 0$$

be the declared observation and weight, and put

$$G_{\mathrm{prof}} = A^*\Omega A.$$

Let the fixed-shift functional be

$$\mathrm{ev}_{h_0}(f) = \langle b_0, f\rangle.$$

Theorem 5.1 (Sharp finite fixed-shift evaluation)**.** *There is a finite K such that*

$$|\langle b_0, f\rangle|^2 \leq K\langle Af, \Omega Af\rangle \qquad (f \in V_X)$$

if and only if

$$b_0 \in \mathrm{ran}(G_{\mathrm{prof}}).$$

When finite, the sharp constant is

$$\boxed{K_X(h_0) = b_0^* G_{\mathrm{prof}}^{\dagger} b_0.}$$

Proof. Diagonalize the positive semidefinite matrix G_{prof}. If $b_0 \notin \mathrm{ran}(G_{\mathrm{prof}})$, its projection onto $\ker(G_{\mathrm{prof}}) = \ker(A)$ is nonzero. A corresponding kernel vector has zero observation and nonzero target evaluation, so no finite inequality exists.

If $b_0 \in \mathrm{ran}(G_{\mathrm{prof}})$, write

$$\langle b_0, f\rangle = \langle G_{\mathrm{prof}}^{\dagger/2} b_0, G_{\mathrm{prof}}^{1/2} f\rangle.$$

Cauchy–Schwarz gives the asserted constant. Equality is attained along the nonzero vector $f = G_{\mathrm{prof}}^{\dagger} b_0$ when $b_0 \neq 0$. When $b_0 = 0$, the sharp constant is $K = 0$ and the target functional vanishes identically, so no nonzero extremizer is needed. Thus the constant is sharp [1, 3]. □

Corollary 5.2 (Exact exponent transfer)**.** *Assume the actual averaged profile satisfies*

$$\|Af_X\|_\Omega \leq X^{1-\sigma+o(1)}$$

and

$$K_X(h_0) \leq X^{2\lambda_{\mathrm{loc}}+o(1)}.$$

Then

$$|f_X(h_0)| \leq X^{1-(\sigma-\lambda_{\mathrm{loc}})+o(1)}.$$

This certificate gives a positive fixed-shift power saving only when $\sigma > \lambda_{\mathrm{loc}}$.

Proof. Take square roots in theorem 5.1 and insert the two assumptions. □

Remark 5.3 (Squared versus amplitude cost). K_X is a squared evaluation cost. The amplitude ledger charges $\sqrt{K_X}$, hence λ_{loc}, not $2\lambda_{\text{loc}}$. The actual endpoint registry must record this convention exactly.

6 Direct slicing is not averaged recovery

Proposition 6.1 (An average can erase fixed evaluation)**.** *Two profiles can have the same average and different values at h_0.*

Proof. On two shifts, let $A(x_1, x_2) = (x_1 + x_2)/2$ and evaluate the first coordinate. The profiles $(1, -1)$ and $(0, 0)$ have the same average but different first coordinates. Equivalently, $(1, -1) \in \ker(A)$ is a witness that the evaluation vector is not in $\operatorname{ran}(A^*A)$. □

Thus a proof of corollary 3.2 uses no Christoffel inversion, whereas a proof based only on Af_X must invoke theorem 5.1. Charging both as independent losses for the same map double-counts; using the direct identity to erase a localization loss when only averaged evidence is available is equally invalid.

7 Relation to the determinant and zero-mode interfaces

TPC-121 and TPC-122 deliberately retain literal fixed-h_0 coefficients [6, 7]. Their conditional outputs are, schematically,

$$\lambda_D^{\text{cert}} \quad \text{and} \quad \eta_Z^{\text{cert}}.$$

This input type does not prove that a preceding frozen family was de-aliased correctly. To attach either formula after the physical archive, the following must be supplied:

I1. the shift-tagged TPC-123 path archive;

I2. the literal TPC-124 grouping and block reassembly;

I3. either the direct operator identity $\Delta_X = 0$, or the actual profile space and its finite $K_X(h_0)$;

I4. one inherited normalization and no reuse of determinant reserve as localization or physical endpoint slack.

Even after these items, the growing inputs in TPC-121 and TPC-122 remain unproved. In particular, this paper cannot fill H3, H5, H7 or H8 by itself.

8 Verdicts and quantifier ledger

Definition 8.1 (Paper-specific verdicts)**.** GO-INTERFACE means that the relevant coefficientwise operator identity or sharp localization input is proved for the declared growing family. STOP-DECLARED-ROUTE means an exact composite defect or kernel witness stops the specified direct or averaged route. NOT-TESTABLE means that the literal shift archive, profile space or growing cost is absent.

Target	Required evidence	Verdict rule
Direct archive slice	$\Delta_X = 0$ coefficientwise for the symbolic family	Nonzero $\Delta_X c_X$ is STOP-DECLARED-ROUTE; missing stages are NOT-TESTABLE.
Stagewise proof archive	Every $D_j = 0$	A nonzero D_j stops that local proof, not every composite proof.
Finite de-aliasing	$b_0 \in \operatorname{ran}(G_{\text{prof}})$	Kernel witness is STOP-DECLARED-ROUTE for the selected profile model.
Growing localization	Actual $K_X(h_0)$ bound in one quantifier scope	A finite-X computation alone is NOT-TESTABLE asymptotically.
Fixed-shift saving	Averaged arithmetic theorem and $\sigma > \lambda_{\text{loc}}$	The interface theorem alone is not L2.

The current TPC-123/124 outputs are exact schemas and finite regression models, not the complete growing shift-tagged archive. The actual profile space required by TPC-115 is likewise absent. Therefore

$$\boxed{\text{FixedShift}_{125} = \text{NOT-TESTABLE from current artifacts.}}$$

This does not show that localization is impossible or that the direct slice must fail.

9 Reproducible audit and claim boundary

The standard-library script `experiments/tpc125_shift_intertwining_audit.py` uses exact rational matrices. It checks two commuting stages, theorem 3.1, a cross-shift leakage defect, a finite localization case, an infinite-cost kernel witness, and the amplitude exponent convention. It is a regression test for the finite theorems, not the missing TPC profile.

Statement	Status
Intertwining telescope and leakage witness	Exact L0.
Sharp localization and exponent transfer	Exact L0.
Attachment to the TPC-123/124 archive	Conditional L1.
Complete growing shift-tagged archive/profile	Absent; NOT-TESTABLE.
Growing averaged or fixed-h_0 arithmetic saving	Not proved; no L2 event.

Explicit exclusions. This paper proves no low-dimensional model for the actual shift profile, no growing Christoffel bound, no signed affine Möbius estimate of TPC-108, no determinant or zero-mode exponent, no physical endpoint below 1/400, no parity breakthrough, no Hardy–Littlewood asymptotic, and no prime-pair or twin-prime theorem [2, 5].

References

[1] Adi Ben-Israel and Thomas N. E. Greville. *Generalized Inverses: Theory and Applications*. Springer, 2 edition, 2003.

[2] Liang Wang. Literal generic-affine mobius dispersion, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-108-literal-generic-affine-mobius-dispersion. TPC-108.

[3] Liang Wang. Fixed-h0 de-aliasing after frozen averaging, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-115-fixed-shift-dealiasing-localization. TPC-115.

[4] Liang Wang. Original hard-packet reconnection, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-119-original-hard-packet-reconnection. TPC-119.

[5] Liang Wang. Tpc-mvp2: A typed global route decision after tpc-103–119, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-120-mvp2-global-route-decision. TPC-120.

[6] Liang Wang. A fiber phase-coherence certificate for post-bin determinant energy, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-121-fiber-phase-coherence-certificate. TPC-121.

[7] Liang Wang. Signed prefixes and the distinguished zero mode, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-122-signed-prefix-zero-mode-transfer. TPC-122.

[8] Liang Wang. Literal native-atom path archives for the tpc-15 hard packet, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-123-literal-native-atom-path-archive. TPC-123.

[9] Liang Wang. Provenance-faithful physical reassembly of the tpc hard packet, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-124-provenance-faithful-physical-reassembly. TPC-124.